

Milestone 2 – Feature Engineering & Data Wrangling

1. Introduction

Milestone 2 of the Azure Usage Forecasting project focuses on transforming the cleaned dataset into a structured and enriched format suitable for predictive modeling. Feature engineering is a critical phase in any data science workflow because it enhances the predictive capability of models by extracting meaningful information from raw variables. In time-series forecasting, engineered features help capture seasonality, trends, cyclical behavior, and historical dependencies that are essential for accurate prediction.

The dataset obtained from Milestone 1 was already cleaned and preprocessed. However, it lacked time-aware and lag-based features required for forecasting Azure resource usage trends. Therefore, this milestone emphasizes deriving new features, restructuring the dataset, and ensuring consistency in time granularity.

2. Objective

The primary objective of Milestone 2 is to prepare the dataset for model development by applying feature engineering and data wrangling techniques. The specific goals include:

- Creating time-based features such as Year, Month, and Quarter.
- Generating lag variables to capture historical patterns.
- Introducing rolling statistics to smooth short-term fluctuations.
- Handling missing values introduced during transformation.
- Producing a final, model-ready dataset for forecasting.

3. Dataset Description

The dataset used in this milestone is the cleaned economic dataset from Milestone 1. It contains numerical economic indicators observed over a continuous monthly timeline. These indicators influence Azure usage demand and are suitable for time-series forecasting. Before feature engineering, the dataset primarily consisted of raw numerical values without explicit temporal attributes or derived statistical features.

4. Feature Engineering Process

The following feature engineering steps were implemented to enhance the dataset:

- A Date column was generated using a monthly frequency to establish time indexing.
- Year, Month, and Quarter features were extracted from the Date column to capture seasonality.

- The dataset was sorted chronologically to preserve temporal order.
- Numerical columns were identified for transformation and lag generation.
- Lag-1 features were created to capture dependency on previous month values.
- Rolling mean features were computed to identify underlying trends and reduce noise.
- Missing values caused by shifting operations were handled using backward filling.

5. Data Wrangling Techniques

Data wrangling ensured the dataset maintained structural consistency and model compatibility. Column names were standardized, numerical columns were verified, and time granularity was maintained at a monthly level. Special care was taken to handle missing values introduced by lag and rolling operations to prevent distortion in forecasting results. All transformations were validated to ensure no unintended data leakage occurred.

6. Output Dataset

The final transformed dataset was exported as 'economic_data_feature_engineered.csv'. This dataset now includes original economic indicators along with derived time-based and lag-based features. The enriched dataset is fully prepared for machine learning and time-series forecasting models in Milestone 3.

7. Tools and Technologies Used

- Python Programming Language
- Pandas Library for data manipulation and feature creation
- NumPy for numerical computations
- Google Colab for development and execution
- GitHub for version control and project management

8. Conclusion

Milestone 2 successfully transformed the cleaned economic dataset into a feature-rich, structured, and model-ready format. By incorporating time-based attributes, lag variables, and rolling statistics, the dataset now effectively captures trend and seasonal patterns. These enhancements significantly improve the forecasting potential of the Azure Usage Prediction system. The project is now ready to proceed to Milestone 3, which focuses on model development, evaluation, and performance optimization.